

PAES: PRODUCTION MACHINERY

PAES 101: 2000

Agricultural Machinery – Technical Means for Ensuring Safety - General

Prevention of accidents arising from the use of tractors and machinery for agriculture and technical means of improving the degree of safety of operators and others.

All moving parts shall be treated as dangerous.

GUARD TYPES

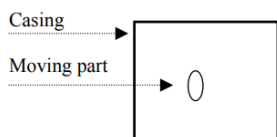
Shield or Cover

Prevent the dangerous part being reached from the **side or sides covered**.



Casing

Prevents contact with the dangerous part from **all sides**



Enclosure

which by means of a rail, fence, frame or the like ensures the safety, distance necessary so that the dangerous part cannot be reached inadvertently.

*Guards shall be sufficiently strong, unless stated, withstand a perpendicular static load of **1,200 N**.*

*Safety distance for upward reach is **2,500 mm** for persons standing upright.*

*Barriers, the height of which is **less than 1,000 mm** above the location, which a person can occupy, shall not be acceptable.*

Operating Positions

A foot-guard (toe-board) on all side shall not be more than **50 mm** farther away and not less than **75 mm** above the platform.

A guard-rail shall not be less than **1000 mm** and not more **1100 mm** above the platform and two rails does not exceed **500 mm**.

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Agricultural Machinery – Operator's Manual – Content and Presentation

Left-hand side

Side which is on the left when an observer is facing the machine

Right-hand side

Side which is on the right when an observer is facing the machine

Publication of operator's manual

- the name and address of the manufacturer and/or distributor of the machine;
- the name and address of the importer of the machine (if imported);
- the model designation of the machine;
- the name or type of publication;
- the part number or publication number by which the manual may be ordered;
- the printing or publication date;
- the language in which the manual is written.

Paper Size

A5 (210 mm x 148 mm) format is suitable for most cases

1/3 A4 format is adequate for simpler machinery and equipment.

A4 (297 mm x 210 mm) format is recommended for complex machines and static equipment.

Instructions

WARNING and CAUTION: safety related information

IMPORTANT: instructions when machine damage is involved

NOTE: supplementary information

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Agricultural Machinery – Method of Sampling

Acceptance test

Test on samples selected from a lot for the purpose of acceptance of the lot

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Lot

All components of equipment under study

same kind, type, size and manufactures from the same material shall be group together

Routine test

Check the specifications which are likely to vary during production

Type test

Prove conformity to the requirements of the relevant specifications

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Agricultural Machinery – Location and Method of Operation of Operator's Controls – Control for Agricultural Tractors and Machinery

Agricultural tractor

Self-propelled, wheeled, track-laying designed to pull, push, and/or operate trailers or provide power to implements and machines

Pedestrian-operated machine

Having an integral power unit but normally operated by a pedestrian

Right-hand and left-hand

Designations related to the operator when sitting on the operator's station

Self-propelled machine

Having one or more integral power units which propel and operate the machine

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Agricultural Machinery – Symbols for Operator's Controls and Other Displays – Common Symbols

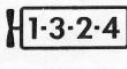
Agricultural machines

Tractors, self-propelled and pedestrian-operated machines, implements, and combinations

Symbol

Visually perceptible figure used to transmit information independent of language

drawing, printing, letters and numerals or other

 Engine R.P.M.	 Engine Oil Pressure	 Firing Order	 Transmission
 Transmission Oil Pressure	 Transmission Oil Temperature	 Transmission Oil Filter	 Neutral
 Park	 Lowered	 Raised	 Rock Shaft
 Connect/Disconnect	 Axle Disconnect	 Reverse	 Differential Lock
 On/Off Power Take-off	 Engaged/Disengaged Hand Brake	 Tow	 Forward
 Reverse	 Control Lever Operating Direction	 Down/Up Vertical Movement	 Engage or In/Disengage or Out Connection
 Extended/Retracted Remote Cylinder	 Raise/Lower	 Dump	 Roll Back
 Bucket Lift Up and Down	 Header	 Reel Height	 Platform Height
 Reel Speed	 Cylinder Speed (Combine)	 Concave Clearance (Combine)	 Full
 Ground Speed	 All Mechanisms	 Fuel	 1/2
 Oil Level	 Dipstick	 Oil Lubricant Type & Frequency	 Grease Lubricant Frequency
 Engine Coolant Temperature	 Air Filter	 Pressurized Open Slowly	 Upper Beam/Lower Beam Lights
 Turn Signals	 Horn	 Windshield Defroster	 Windshield Wiper
 Windshield Washer	 Windshield Wiper and Washer	 Lighter	 Ammeter or Generator Light
 Temperature	 Temperature Control	 Ventilating Fan	 Speed Range

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 Continuously Variable (Speed Range)	 Hours	 Fasten Seat Belt	 Caution	 Read Operator's Manual
 Heavy - - - Light	 Increase	 Decrease		

Used on illuminated displays

RED: failure or serious malfunction; requires immediate attention

YELLOW or AMBER: outside normal operating limits

GREEN: normal operating condition

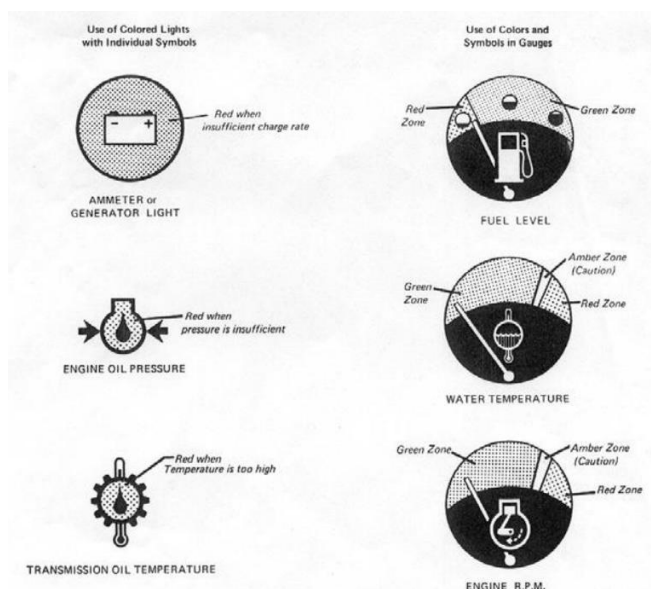
Used for specific functions

BLUE: headlight main- / high-beam display;

RED: hazard warning display; and

GREEN: turn signal display

RED for hot and **BLUE** for cold



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Agricultural Machinery – Soil Tillage and Equipment - Terminology

A BASIC TILLAGE GOALS

Tillage action

Action of a tillage tool in executing a specific form of soil manipulation

Ex. soil cutting, shattering/ inversion

Tillage objective

Desired soil condition produced by one or more tillage operations

Tillage requirement

Soil physical conditions which can be produced by tillage and is necessary based on utilitarian and/or economic considerations

B GENERAL TILLAGE TERMS AND KIND OF TILLAGE

Broadcast tillage

Overall tillage

Tillage of an entire area as contrasted to a partial tillage as in bands or strips.

Deep tillage

Primary tillage operation which manipulates soil to a greater depth than 300 mm

accomplished with heavy-duty moldboard or disc plow

Earthmoving

Tillage action and transport operations utilized to loosen, load, carry, and unload soil

Land forming

Move soil to create desired soil configurations

large scale such as contouring or terracing, small scale such as ridging or pitting

Land grading

Move soil to establish a desired soil elevation and slope

Ex. Leveling, contouring, cutting, and filling

Land planing

Cuts and moves small layers of soil to provide smooth, refined surface condition

Oriented tillage

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Oriented in specific paths or directions with respect to the sun, prevailing winds, previous tillage actions, or field base lines

Rotary tillage

Employing power-driven rotary action to cut, break up and mix soil

Soil cultivation

Shallow tillage operation performed to promote growth of crop plants by creating a soil condition conducive to aeration, infiltration, and moisture conservation or to pest control

Tillage

Mechanical manipulation of soil for any desired purpose

changing of soil conditions for the enhancement of crop production

Primary tillage

Initial major soil-working operation, designed to reduce soil strength, cover plant materials, and rearrange aggregates

Secondary tillage

Following primary tillage, to control weed growth and to create specific soil surface configurations before seeding

Tillage depth

Tool depth

Vertical distance from the initial soil surface to a specified point of penetration of the tool

C

TILLAGE SYSTEMS

Conservation tillage

Maintains a minimum of 30% residue cover on the soil surface after planting or maintains at least 1,100 kg/ha of flat small grain residue equivalent on the soil surface during the critical erosion period

Conventional tillage

System traditionally performed in preparing a seedbed for a given crop and grown in a given geographical area

Minimum tillage

Least soil manipulation is performed

Mulch tillage

Tillage of the total soil surface is performed in such a way that plant residue is specifically left on or near the soil surface

Optimum tillage

Idealized system which permits a maximized net return for a given crop under given conditions

Precision tillage

Subsoiling under the plant row prior to planting usually intended for subsurface drainage

Reduced tillage

Primary tillage operation is performed in conjunction with special planting procedures in order to reduce or eliminate secondary tillage operations

Reservoir tillage

Large number of depressions or small reservoirs are formed to hold rain or sprinkler applied water

Ridge tillage

Ridges are formed during cultivation or after harvest and maintained from year to year in the same location

seeding is done on the ridge top

V tillage

Only isolated bands of soil is tilled

D

SPECIFIC TILLAGE OPERATION

Anchoring

Tillage to partially bury and thereby prevent movement of materials such as plant residues or artificial mulches

Bedding

Ridging

Listing

Forms a ridge and furrow soil configuration

Bulldozing

Pushing or rolling of soil by a steeply inclined blade

Chisel plowing

Narrow curved shank is used

depths greater than 350 mm is termed as subsoiling

Combined tillage operations

Simultaneously utilizing two or more different types of tillage tools or implements to simplify, control, or reduce the number of trips over a field

Harrowing

Pulverizes, smoothens, and makes the soil ready for planting

used before seeding

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Incorporating

Mixing

Operation which mix or disperse foreign materials, such as pesticides, fertilizers or plant residues into the soil

Middlebreaking

Hilling-up

Lister is used in a manner that forms a furrow midway between two previous rows of plants

Off barring

Cuts and throws the soil away from the base of plants

reverse of middlebreaking or hilling-up

Moldboard plowing

Cut the soil with partial or complete soil inversion

Residue processing

Cut, crush, ancho, or otherwise handle residues in conjunction with soil manipulation

Subsoiling

Deep tillage, below 350 mm for the purpose of loosening soil for root growth and/or water movement

Vertical mulching

Vertical band of mulching material is injected into the slit immediately behind a tillage tool shank

E

TILLAGE EQUIPMENT

GENERAL-PURPOSE TILLAGE IMPLEMENT

Perform functions simultaneously that of initial cutting, breaking and pulverizing the soil

Plow-harrow

Works under the combined principles of the regular disc plow and harrow

has a frame, wheel arrangement and depth adjustment

Rotary tiller

Used for broadcast or strip tillage and also used as chemical incorporator and arow crop cultivator

consists of power-driven shaft, transverse to the direction of travel with curved knives

Spiral plow

Rotary plow

Consists of two horizontal power driven spiral flanged shafts which rotate vertically

placed end-to-end and oriented to throw the soil outward

PRIMARY TILLAGE IMPLEMENT

Used for cutting, displacing and/or shattering the soil to reduce soil strength and to bury or mix plant materials, pesticides, and fertilizers

Chisel plow

Shatters the soil without complete burial or mixing of surface materials

Disc plow

With individually mounted concave disc blades which cut, partially or completely invert a layer of soil to bury surface material, and pulverize the soil

Moldboard plow

Cuts, partially or completely inverts a layer of soil to bury surface materials and pulverizes the soil

Right-hand plow

Turns the furrow slice to the right of the plow

Left-hand plow

Turns the furrow slice to the left of the plow

Two-way plow

Eliminates back and dead furrows and is used to surface irrigation

consists of both right-hand and left-hand plows, with one type being used at a time

SECONDARY TILLAGE EQUIPMENT

Used for tilling the soil to a shallower depth than primary tillage implements, additional pulverization, mix pesticides and fertilizers into the soil, level and firm the soil, close air pockets, and eradicate weeds.

Comb-tooth harrow

Breaking clods after initial plowing, for subsequent operations prior to transplanting and for puddling and leveling

works like a rake

Disc harrow

Used to pulverize the soil to attain a better soil tilth for the seed germination and growth

two or four gangs of concave steel discs

Single-action disc harrow

Consists of two gangs of disc, placed end-to-end at an angle, which throw the soil in opposite directions

Double-action disc harrow

Tandem disc harrow

Consists of two or more gangs, in which a set of two gangs follows behind the from

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gangs and is arranged in such a way that the discs on the front gangs throw the soil in one direction (usually outward) and the discs on the rear gangs throw the soil in the opposite directions

Offset disc harrow

Consists of two gangs wherein one gang is located behind the other at an angle and the harrow is operated in an offset position in relation to the tractor

Field cultivator

Implement for seedbed preparation, weed eradication, or fallow cultivation subsequent to some form of primary tillage

**with spring steel shanks or teeth (spaced 150-230 mm in staggered)*

Packer

Crushing soil clods and compacting the soil

one or two in-line gangs of rollers

Roller-harrow

Used for seedbed preparation which crushes soil clods and smooths and firms the soil surface

consists of in-line gang of ridged rollers, followed by one or more rows of staggered spring cultivator teeth, followed by second in-line gang

Rotary hoe

Dislodging small weeds and grasses and for breaking soil crust and is used for fast, shallow cultivation before or soon after crop plants emerge

Row crop cultivator

The frame and cultivating tools are designed to adequately pass through standing crop rows without crop damage

Spike-tooth harrow

Consists of long spike attached rigidly to cross bars and staggered to attain maximum stirring and raking of soil

Spring-tooth harrow

Consists of long, flat and curved teeth made of spring steel

CULTIVATING TILLAGE IMPLEMENT

Shallow post-plant tillage to aid the crop by loosening the soil and/or by mechanical eradication of undesired vegetation

Continuous-tool bar cultivator

Consists tool bars that extend across the top of the rows, which allow lateral adjustments of the tools for different row spacing

Separated gang cultivator

Consists of tool bars that drop down between the rows to provide maximum vertical clearance for the plants

F

NOMENCLATURE FOR TILLAGE TOOLS AND IMPLEMENTS

Bed shaper

Soil-handling which forms uniform ridges of soil to predetermined shapes

Blade

Consists of an edge and a surface, cut through the soil

rotary tiller blades, anhydrous ammonia blades

Coulter

cut plant material and soil

Draft

force to propel an implement in the direction of travel which is equal and opposite to drawbar pull

Effective operating width

operating width excluding overlap

Edge clearance angle

effective angle which is included between the line of travel and a line drawn through the back or nonsoil-working surface of the tool and its immediate edge

Ground clearance

minimum vertical distance between the soil surface and a potentially obstructing machine element

Hitch

portion of an implement designed to connect the implement to a power source

Implement width

Horizontal distance perpendicular to the direction of travel between the outermost edges of the implement

Injector

Insert materials into the soil

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Jointer

Miniature plow attachment which turn over a small furrow slice directly ahead of the main moldboard plow bottom, to aid in covering trash

Lateral tool spacing

Horizontal distance between corresponding reference points on adjacent tools when projected upon a vertical plane perpendicular to the direction of travel

Line of travel

Line and direction along which the tillage implement travel

Lister-planter

Lister and planting combined tillage implement to permit single listing-seeding operation

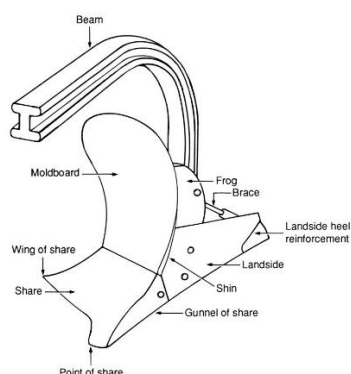
Longitudinal tool spacing

horizontal distance between corresponding reference points of two tools when projected upon a vertical plane parallel to the direction of travel

mechanical tillage implement

single or groups of soil-working tools together with power transmission structure, control, and protection systems present as an integral part of the machine

G MOLDBOARD PLOW CLEARANCES



Horizontal clearance

distance measured between specified points on adjacent plow bottoms

Ex. diagonal (rake), tip of share to tip of share; fore and aft, width of cut or furrow slice; throat width, minimum distance from face of moldboard to projecting member of preceding bottom

Vertical clearance

distance measured from cutting edge of share to nearest potentially obstructing member such as main truss (backbone), frame, beam, release mechanism, etc

Operating overlap

distance perpendicular to the direction of travel that an implement reworks soil previously tilled

Operating width

horizontal distance perpendicular to the direction of travel within which an implement performs its intended function

Protected zone

soil and/or plant zone purposely protected by virtue of tool design, tool spacing or evasive tool movement

Scouring shedding

soil-tool reaction in which soil slides over the surface of the tillage tool without significant adhesion

Shank

structural member primarily used for attaching a tillage tool to a beam or a standard

Shovel

spade-shaped, V-pointed soil working tool, which is used for various plowstocks, cultivators, grain drills, and soil scarifiers

Side force

side draft

horizontal component of pull, perpendicular to the line of motion

Soil-additive applicator

machine used to apply, or to apply and incorporate soil additives by means of tillage

Ex. Granular herbicide applicator, lime or manure spreader, fumigation and fertilizer distributor, or chemical incorporator are examples of soil-additive applicator.

Soil-additive incorporator

machine used to mechanically incorporate or mix material into the soil

Soil opener

tillage tool used to slice through soil and create an opening for the insertion of material such as seeds, pesticides, fertilizers

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Ex. disc, knife, and runner

Soil roller

rotating implement which pulverizes, firms or smooths soil by crushing or compacting

Soil-sliding path

path along which one element of soil slides across a tillage tool

Soil-sliding path length

length of the path along tillage tool upon which soil slides

Soil-ascending angle

angle between the sliding path and the horizontal at any point along the sliding path

Soil-sliding angle

angle at any point on the surface of a tool between the soil sliding path and a horizontal contour line constructed through the surface of the tool

Soil-tool geometry

configuration of the soil-tool boundary wherein the overall shape is usually oriented with the direction of travel of the tool and the soil surface

Soil-working surface

portions of tillage tools which are designed to be in contact with the soil

Specific draft

unit draft

draft force of an implement per unit area of tilled cross-section

Standard

beam

upright support which connects the shank to tillage implement frame

Sweep

type of cultivator shovel which is wing-shaped

Teeth

projections on tillage tools which serve to penetrate, grip, cut, or tear soil

Tillage tool

individual soil-working element

Complex tillage tools

rotate or move so that they present a varying boundary and contact area to the soil

Ex. clod breakers, notched discs, rotary hoes

Dynamic tillage tools

powered so that some of their movements are in direction other than along the line of travel

Multi-powered tillage tools

powered by more than one form of power, such as draft and rotating power, or draft and electrical power

Simple tillage tools

present a reasonable constant boundary area to the soil

Tool clearance

Minimum distance in a specified direction between a point on the tool and the nearest potentially obstructing implement element

Tool-operating width

Maximum horizontal distance perpendicular to the line of motion over which a tool perform its intended function

Orientation, tool

Position of the tool in a framework of cartesian coordinates which is usually oriented with the soil surface and the direction of travel

orientation is specified in side, tilt, and lift angles as a minimum

Lift angle

Rake angle

Angle, in a vertical plane parallel to the direction of travel, between a tool axis and the soil surface

Side angle

Angle, in the soil surface plane, between a tool axis and a line, which is perpendicular to the direction of travel

Tilt angle

angle, in a vertical plane perpendicular to the direction of travel, between a tool axis and the soil surface

Tool overlap

distance perpendicular to the direction of travel in which a tool operating width coincides with the operating width of another tool

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tool-skip area

area of soil surface left undisturbed during passage of a tool

tool width

maximum horizontal projection of a tool in the soil perpendicular to the line of motion

vertical tool spacing

vertical distance between corresponding points on adjacent tools when projected upon a vertical plane parallel to the direction of travel

wings

projections attached to the sides of tillage tools to increase the volume of soil which can be disturbed, or to control the nature and distance of soil movement. Wings usually have lift, tilt, and side angles which are different from those found in the orientation of the main tool and standard

H

SOIL REACTION NOMENCLATURE

soil abrasion

scratching, cutting, or abrading of materials caused by the action of soil

soil adhesion

sticking of soil to objects such as tillage tools or wheels

soil compaction

act of reducing the specific volume of soil

soil cutting

separation of a soil mass by a slicing action

soil failure

alteration or destruction of a soil structural condition by mechanical forces such as in shearing, compression, or tearing

soil heaving

lifting or swelling of soil resulting from natural forces such as freezing

soil reaction

soil response to the application of mechanical forces

soil shatter

pulverization
general fragmentation of a soil mass resulting from the action of tillage forces

soil sliding

sliding of soil across a surface

throw

movement of soil in any direction as a result of kinetic energy imparted to the soil by the tillage tool

I

SOIL NOMENCLATURE

additive, soil

foreign materials, other than seeds, which are added to and/or incorporated in soil for directly influencing the soil condition or environment

Ex. pesticides, fertilizers, mulches, or conditioners, but not foreign bodies such as drain tiles, which have an indirect influence

adhered soil bodies

masses of soil (may be stationary or in a relatively slow motion) which adheres on soil working surfaces and act as a part of the tool

Ex. soil cone, an adhered soil body which resembles a cone; soil sheet, an adhered soil body which covers a large area of tool like a sheet; soil wedge, an adhered soil body which resembles a narrow wedge

Compacted layer

hard pan

plow pan

plow soil

dense layer of soil immediately below tillage depth created by mechanical pressure and/or soil-shearing forces

concretions

soil structural units which are irreversibly cemented together

covering depth

thickness of soil with which materials are covered by an implement

foreign materials

all materials added to or mixed into soil, including residues, soil additives, and foreign bodies that have not originated in the soil's development

mechanical impedance

resistance to the movement of plant parts or tillage tools through soil that is caused by the mechanical strength of the soil

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mechanical stability

mechanical strength

degree of resistance of soil to deformation

shear blocks

clods

blocks of soil which are sheared loose from the main soil mass by tillage tool action

shear surface

failure surfaces occurring where the soil has sheared

primary shear surfaces

initial and distinct surfaces appearing during failure which are caused mainly by shear

secondary shear surfaces

shear surfaces which result from the twisting, pushing, or tumbling of the soil after or during the initial displacement

Secondary shear surfaces are often perpendicular to the primary shear surfaces

soil aggregates

soil peds

agglomerations of primary soil particles which are produced by natural processes

tillability

degree of ease with which a soil may be manipulated for a specific purpose

J

SOIL AND SURFACE CHARACTERISTICS

Back furrow

Raised ridge left at the center of the strip of land, when plowing is started from center to side

It is formed when a furrow slice is lapped over another slice

Dead furrow

open trench (about twice the width of one plow bottom) left in between the adjacent strips of land after finishing of plowing

It is formed when two adjacent furrow slices are thrown opposite each other

Furrow

Trench left when the plow bottom cuts and turns the furrow slice

Furrow crown

Peak of the turned furrow slice

Furrow depth

ditch depth

pit depth

trench depth

depth of depression below a specified (initial or subsequent) soil surface

furrow slice

soil mass cut, lifted, pulverized, inverted and thrown to one side of the plow bottom

furrow wall

undisturbed or unbroken side of the furrow

head land

unplowed soil at the end of the furrow strip

land

unplowed soil

ridge height

bed height

hill height

windrow height

height of soil above a specified (initial or subsequent) soil surface

root bed

soil profile modified by tillage or amendments for use by plant roots

root zone

part of the soil profile exploited by the roots of plants

seedbed

soil zone which affects germination and emergence of seeds

soil density

weight of a unit volume of soil expressed on either a wet basis (including soil and water) or on a dry basis (soil only, most common)

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Agricultural Machinery – Hitch for Walking-type Agricultural Tractor - Specifications